HRISHIKESH SINGH

PhD and PMRF Doctoral Research Scholar
Department of Water Resources Development and Management, Indian Institute of Technology Roorkee, India

☎ +91-9753391666 ✉ hrishiksh_s@wr.iitr.ac.in ; hrishikesh618@gmail.com

Hydrodynamic Modelling | Flood Risk & Vulnerability | Hydroclimatic Extremes

EDUCATION

Doctor of Philosophy **July 2022 - Present**
Indian Institute of Technology Roorkee, India
Dept. of Water Resources Development and Management

Master of Technology in Water Resource Dev. and Irrig. Engg. **2020 - 2022**
Dept. of Civil Engineering **(**Silver Medallist**)**
National Institute of Technology Raipur, India

Bachelor of Engineering (B.E.) **2015 - 2019**
Department of Civil Engineering
Shri Shankaracharya Technical Campus

RESEARCH INTERESTS

- Global and Regional Flood models, Numerical 1D, 2D and coupled Hydrodynamic modelling, Hazard and Risk mapping
- Socio-economic and Infrastructure Vulnerability Analysis, Multi criteria decision making (MCDM), Objective weightage using Entropy, and TOPSIS
- Hydrological Modelling, Extreme Value Analysis, Climate Extremes and indices**,** Wet and Dry Extremes Statistical downscaling, Drought and Climatic Hazard
- Design Rainfall analysis for Data-scarce ungauged/partially gauged catchments, Multiday, Design Hyetograph and Hydrograph production using statistical parametric and non-parametric distributions
- Geospatial and Satellite-imagery analysis, Utilising Satellite data for Hydroclimatic applications, Supervised LULC classification, Satellite-based Floodplain Mapping

TECHNICAL SKILLS

- **Hydrodynamic Models**: LISFLOOD-FP, HEC-RAS; **Hydraulic Model**: SWMM; **Hydrological Model**: HEC-HMS
- **GIS & RS Analytics**: Google Earth Engine, GDAL, ArcGIS, QGIS
- **Big Data & HPC**: Parallel computing, Linux environments, Large-scale data processing and workflow automation.
- **Programming***:* Python and its libraries for Data handling, *R* and *MATLAB*
- **Generative AI**: Prompt engineering for LLM-assisted code development and data analysis
- **Structural design packages**: AUTOCAD
- **Image Designing and Editing**: Canva and Adobe Photoshop

LANGUAGE

- English: C1 (reading), C1 (writing), C1 (listening)
- Hindi: Native
- Sanskrit: Elementary (A2)

PUBLICATIONS

- **Singh, H**., & Mohanty, M. P.* (2023). Can Atmospheric Reanalysis Datasets Reproduce Flood Inundation at Regional Scales? A Systematic Analysis with ERA5 over Mahanadi River Basin, India, ***Environmental Monitoring and Assessment*, Springer. https://doi.org/10.1007/s10661-023-11798-2 (IF -3.0)**
- **Singh, H**., Mohanty, M.P. Performance of ERA5 Products in Capturing Pluvial and Fluvial Inundation: A Statistical-Cum-Hydrodynamic Framework for Resource-Constrained Large Flood-Prone Watersheds. Water Resources Management (2025). https://doi.org/10.1007/s11269-025-04257-9 **(IF -4.7)**
- **Singh, H.**, & Mohanty, M. P. (2025). Multi-scale assessment and entropy-MCDM framework for evaluating reanalysis precipitation datasets over Indian basins. International Journal of Applied Earth Observation and Geoinformation, 144, 104919. https://doi.org/10.1016/j.jag.2025.104919 **(IF -8.6)**
- Tyagi, M., **Singh, H**., Thakur, D. A., & Mohanty, M. P. (2024). Compound risks of floods and droughts over multi-hazard catchments: Revealing association through hydrodynamic-cum-statistical modelling and novel bivariate risk classifier. Science of The Total Environment, 957, 177689. https://doi.org/10.1016/j.scitotenv.2024.177689 **(IF -8.0)**
- Mishra, D., **Singh, H**., Kumar, M., & Mohanty, M. P. (2025). Advanced hydrological assessment with SWAT+ under climate change: An integrated framework over large climate-sensitive and reservoir-regulated watershed. Science of The Total Environment, 994, 180062. https://doi.org/10.1016/j.scitotenv.2025.180062 **(IF -8.0)**
- Deopa, R., **Singh, H**., Tripathi, V. et al. Cascading impacts of 2025 June-July multi-hazard episodes in the Western Himalayas: Evidence through remote sensing observations and Google Earth Engine. Nat Hazards 121, 23417–23422 (2025). https://doi.org/10.1007/s11069-025-07699-x (**IF -4.5**)

CONFERENCES

- **20 conference presentations** (11 as first author) at international and national conferences, including AGU, EGU, JpGU–AGU, IAHS, WCRP Open Science Conference, ICRC-CORDEX, ICR6 and EWRA. The complete list is provided in **Appendix A**.

TEACHING AND MENTORSHIP

Teaching Assistantships

Drinking Water, Sanitation and Sustainability **Autumn 2023, 2024, 2025**
M.Tech. (Drinking Water and Sanitation), Dept. of Water Resources Development and Management, IIT Roorkee

Design of Irrigation and Drainage Works **Autumn 2023, 2024**
M.Tech. (Irrigation Water Management), Dept. of Water Resources Development and Management, IIT Roorkee

River Engineering **Spring 2024, 2025**
M.Tech. (Water Resources Development and Irrigation Water Management), Dept. of Water Resources Development and Management, IIT Roorkee

Fluid Mechanics I and Fluid Mechanics II **2021 - 2022**
B.Tech. (Civil Engineering), Dept. of Civil Engineering, NIT Raipur

Training Programmes and Lectures Delivered

Training Programme on Hydrological and Flood Modelling (3-day, hands-on) **16 - 18 Oct 2025**
Larsen & Toubro (L&T), Chennai
River morphology, hydrodynamic modelling, HEC-HMS, flood frequency analysis and hazard mapping, HEC-RAS (1D/2D) and urban flood modelling with SWMM

Hands-on Lecture: Analysis and Processing of Precipitation Data **2024**

PMRF Teaching Assistantships (outside IIT Roorkee; 154 contact hours)

Basic Statistics **Aug 2025 - Nov 2025**
B.Sc., Burman Group of Institutions, Roorkee
Offline tutorials, 52 contact hours

Environmental Studies and Disaster Management **Jul 2024 - Jan 2025**
B.A. (Geography), Chaman Lal Mahavidyalaya, Landhaura, Haridwar
Offline tutorials, 52 contact hours

Construction Management, Accounts and Entrepreneurship Development **Feb 2024 - May 2024**
3rd-year Diploma (Civil Engineering), College of Engineering Roorkee (COER)
Offline lectures and tutorials for a class of 17 students, 50 contact hours

Student Mentorship and Supervision

Mentored 1 M.Tech. Dissertation **2024 - 2025**
Dept. of Water Resources Development and Management, IIT Roorkee

Mentored 1 M.Tech. Dissertation **2023 - 2024**
Dept. of Water Resources Development and Management, IIT Roorkee

Supervision of 6 Bachelor Interns
GIS Techniques in Hydrology and Urban Flood Modelling

PROFESSIONAL EXPERIENCE

Project: Climate Adaptation in Wetlands along the Mahanadi River Catchment in Chhattisgarh **Jan 2022 - Mar 2022**

HONORS AND AWARDS

- Received the Prestigious **Prime Minister Research Fellowship**, 2024 in the 11th cycle
- Felicitated by the Director and the Dean of Academic Affairs, IIT Roorkee, at the **Celebration of Publications in High-Impact Journals** for research publications (August 2026)
- Registration Waiver for the **JpGU–AGU Joint Meeting 2026**, Chiba, Japan (May 2026)
- Travel Grant from the **International Centre for Theoretical Physics** (**ICTP**) to present a paper at the ICRC-CORDEX 2023 from September 25-29, 2023 in Trieste, Italy
- Travel Grant for the **IndoML** Workshop organised by the Indian Institute of Technology Bombay
- **Silver medal** for academic excellence in M.Tech. (WRD&IE, NIT Raipur)

FIND ME

Personal Website | Google Scholar | ORCID | ResearchGate | LinkedIn

APPENDIX A: CONFERENCE PRESENTATIONS

- **Singh, H**., Tyagi, M., & Mohanty, M. P.* (2026). Multi-Scale Flood Risk Assessment across India: Copula-Based Urban Compound Extremes, Critical-Duration Basin Flood Response, and Socio-Hydrological Vulnerability. AGU26 Annual Meeting, December 7–11, 2026, San Francisco, California, USA.
- Saha, S., **Singh, H**., & Mohanty, M. P.* (2026). A Computationally Efficient Framework Integrating Remote Sensing and Hydrodynamic Modelling for Escalating GLOF Hazards in the Nepal Himalaya. JpGU–AGU Joint Meeting 2026, Session M-IS24, May 24–29, 2026, Chiba, Japan (oral presentation). (received a registration waiver to present)
- **Singh, H**., & Mohanty, M. P.* (2026). TRIC: A Tier-based Risk Index for Assessing and Ranking Urban Flood Risk Across India's 100 Smart Cities Under Compound Extremes. JpGU–AGU Joint Meeting 2026, Session A-HW38, May 24–29, 2026, Chiba, Japan (oral presentation). (received a registration waiver to present)
- Deopa, R., Mishra, D., Dayal, D., **Singh, H**., & Mohanty, M. P. (2026). Integrated modelling of the flood–hydro-ecological–health nexus under future climate extremes in urban megacities. JpGU–AGU Joint Meeting 2026, May 24–29, 2026, Chiba, Japan. (received a registration waiver to present)
- **Singh, H**., & Mohanty, M. P.* (2026). Towards a dual-scale flood risk assessment in India: copula-based urban extremes, basin-scale design flood simulation, and socio-hydrological vulnerability. EGU General Assembly 2026, May 3–8, 2026, Vienna, Austria (EGU26-1115).
- Saha, S., **Singh, H**., Sharma, N., & Mohanty, M. P.* (2026). Nationwide Multi-Scenario GLOF Hazard Mapping in Nepal Using Remote Sensing and Hydrodynamic Modelling. EGU General Assembly 2026, May 3–8, 2026, Vienna, Austria.
- **Singh, H**., & Mohanty, M. P.* (2025). Suitability of Reanalysis products in capturing Flood Inundation and Hazards over India: Deriving insights through Statistical tests and Numerical Flood Modeling. EGU General Assembly 2025, Vienna, Austria, 27 April–2 May 2025, EGU25-755. https://doi.org/10.5194/egusphere-egu25-755
- **Singh, H**., Tyagi, M., & Mohanty, M. P.* (2025). A Novel High-Resolution Flood Hazard Model for India: Integrating Reanalysis Data with Hydrodynamic Simulations. XIIth IAHS Scientific Assembly 2025, October 05–10, 2025, IIT Roorkee, India.
- Mishra, D., Kumar, M., **Singh, H**., & Mohanty, M. P.* (2025). Impact of Climate Change on Hydrology and Extreme Events in the Mahanadi River Basin: A SWAT+ Approach. XIIth Scientific Assembly of the International Association of Hydrological Sciences 2025, October 05–10, 2025, IIT Roorkee, India.
- Mohanty, M. P.*, Tyagi, M., **Singh, H**., & Thakur, D. A. (2024). A novel Bivariate Risk Classifier to capture Compound Risks from Flood and Droughts over Multi-hazard coastal catchments. Multi-hazard DRR Academy for Young Researchers and Practitioners, October 23–26, 2024, Barcelona, Spain. (received a travel grant to present and convene a session)
- Tyagi, M., **Singh, H**., Thakur, D. A., & Mohanty, M. P. (2024). Enhancing Disaster Resilience in the Mahanadi Basin: A Unified Flood and Drought Risk Assessment Using LISFLOOD-SPEI. Young Leaders' Conclave (YLC), National Institute of Disaster Management, New Delhi, India.
- **Singh, H**., & Mohanty, M. P.* (2024). Adequacy of Regional Climate Models in identifying flood hazards at a regional scale? Assessing their performance by attributing outcomes through global flood models in a data-scarce, flood-prone region in India. International Conference on Flood and Water Resources (ICFWR), IIT Roorkee, India, 2024.
- **Singh, H**., & Mohanty, M. P.* (2024). Suitability of Reanalysis products in capturing Flood Inundation and Hazards over India: Deriving insights through Statistical tests and Numerical Flood Modeling. 6th WCRP International Conference on Reanalysis (ICR6), October 28–November 1, 2024, Tokyo, Japan.

- **Singh, H**., & Mohanty, M. P.* (2023). Performance of Atmospheric Reanalysis Products in Reproducing Flood Hazards over India: Investigating their Performance at the River Basin Scale. AGU Fall Meeting, December 11–15, 2023, San Francisco, USA.
- Thakur, D. A., **Singh, H**., & Mohanty, M. P.* (2023). Fidelity of RCMs in Quantification of Climate Change Impacts on Flood Risk – An exhaustive approach over severely flood-prone coastal catchment. International Conference on Regional Climate ICRC-CORDEX 2023, September 25–29, 2023, Trieste, Italy. (received a student travel grant for presentation in Italy)
- **Singh, H**., & Mohanty, M. P.* (2023). Checking the performance of Regional Climate models in determining Flooding at Basin Scale using Global Hydrodynamic model. International Conference on Regional Climate ICRC-CORDEX 2023, September 25–29, 2023, Trieste, Italy. (received a student travel grant for presentation in Italy)
- Thakur, D. A., **Singh, H**., & Mohanty, M. P.* (2023). An Amalgamated Approach to quantify Climate Change Impacts on Flood Risk – Utilisation of RCMs for Efficient Flood Management at Regional Scale. WCRP Open Science Conference 2023, October 23–27, 2023, Kigali, Rwanda.
- **Singh, H**., & Mohanty, M. P.* (2023). Is ERA5 able to reproduce flood inundation at the regional scale? A systematic analysis using LISFLOOD-FP. 12th World Congress on Water Resources and Environment (EWRA 2023), June 27–July 01, 2023, Thessaloniki, Greece.
- **Singh, H**., Tripathi, V., & Mohanty, M. P.* (2023). Can Regional Climate Models represent flood hazards at regional scales? Attributing performance through global flood models over a data-scarce flood-prone region in India. WCRP Open Science Conference 2023, October 23–27, 2023, Kigali, Rwanda.
- Tripathi, V., **Singh, H**., & Mohanty, M. P.* (2023). Fidelity of Machine Learning models in capturing flood inundation through geomorphic descriptors over Ganga sub-basin, India. 12th World Congress European Water Resources Association (EWRA 2023), June 27–July 01, 2023, Thessaloniki, Greece.